

Assignment 2

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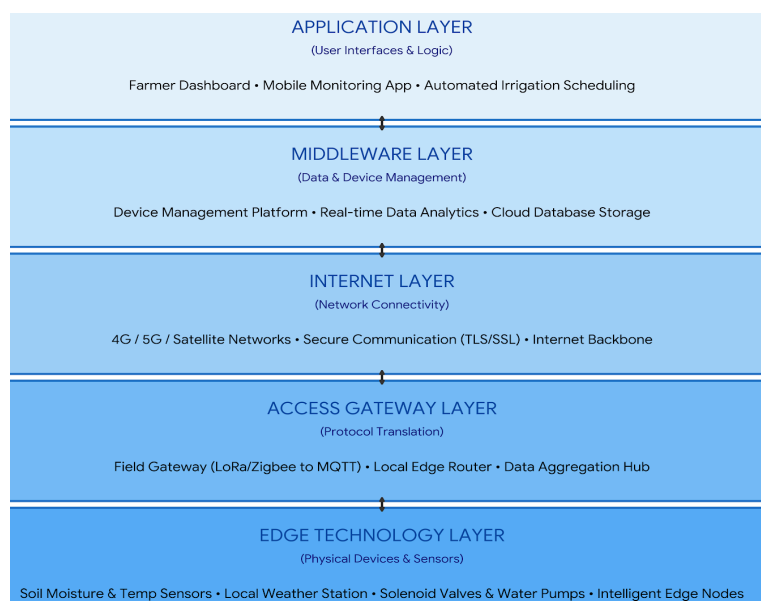
Introduction

An operational technology “OT” is a hardware or software system that monitors or controls physical devices, industrial processes, or equipment in real-world environments. OT is important to our daily lives because it enables the physical processes that provide us our basic needs such as power, water, food, and transportation. Electricity, clean water, and manufacturing would not be given to us without OT and general convenience would not be present.

We can differentiate OT from IT as it focuses on interactions with machinery in industries such as manufacturing, energy, and general utilities. IT focuses on data and business operations while OT manages physical processes and equipment. IT prioritizes confidentiality, integrity, and availability while OT prioritizes safety, reliability, and real-time performance. IT and OT are used in very different environments and have different purposes.

Our Operational Technology

The OT system for our study is an irrigation system for agriculture. The application layer would be a user interface for farmers to see irrigation status, water usage, and any alerts while they can schedule adjustments and generate reports. The middleware layer will handle data processing and storage on cloud platforms that collect sensor data and run irrigation algorithms. The internet layer will connect our



endpoints together using cellular networks, Wi-Fi, and satellites based on location. The access gateway layer will translate IoT protocols to IP-based standards while using edge routers to filter incoming data for actuators. Lastly, the edge technology layer involves sensors and actuators to detect soil moisture, weather, rate of water flow, and functionality of solenoid valves.

Uncovering the Weaknesses - Vulnerability Assessment

Physical Access to the Device: Under optimal conditions only the farmer in question or any employees should have access to the media dashboard used to control the irrigation system, however risks can appear, such as tampering by an employee, or if there is a lack of physical deterrence like fencing around the facility used by the farm, an intruder may tamper with the irrigation dashboard. Other risks also include tampering with networking and cloud infrastructure by remote attackers, if the farm is considered a target. Sensors on the Edge Technology layer can also be tampered with with bad actors if access to them is far too open, such as a lack of storage area for equipment after use, or if there is an insufficient failsafe for sensor calibration. Another situation to note would be what measures are used to secure the perimeter of the facility, such as electrified fencing, or at the very least barbed wire at the top of the fence. If the physical defenses of the facility are easily breached, the equipment will be under severe threat from bad actors.

Data Transmission Security: When it comes to data transmission, it would be optimal that all data collected by the sensors from the Edge Technology Layer and Access Gateway Layer should be encrypted, especially if being sent to a cloud based database. However, if there is a lack of encryption, this opens up the risk of packet sniffing. The security of the cloud based database should also be noted, as if the Cloud Provider has weak security that

will also lead to a breach of the irrigation data being collected. At the internet layer all of our data should be encrypted, any lack of it is an immediate security risk.

User authentication and access control: When it comes to user authentication and access control, only specific workers should have access to the dashboard to the irrigation system or require admin authentication to change specific settings. Employees should be able to change the core function of the irrigation system, or the lead worker of the facility. It should also be implemented that the manager should be the only one aside from IT admins that have access to the cloud database from sensor calibration and data storage, as this would open a threat to disgruntled employees tampering with the system or bad actors outside the facility. Overall, when it comes to authentications and access control, systems should be put in place to validate specific tasks workers may need to handle, and systems should be in place to validate those workers and managers/IT alike.

Default settings and configurations: All default settings and passwords should be changed as soon as possible, as this would make it easy for hackers and disgruntled employees alike to attack the system and change critical functions. Given that irrigation systems are also used in specific scientific experiments for medicine production and crop yields, default settings should be changed as the first order of business, as any attacker after doing simple research on the system can and will exploit any unchanged default settings, which can lead to disastrous results for the operation at hand.

Becoming a Defender

Security measures we can implement to fortify our systems are implementing a strong password policy, applying access controls to restrict access to administrator privileges and files for all users, and protecting the data through encryption.

Password policies can be implemented in an Active Directory environment by creating a Group Policy Object for it that can be applied to all users. This policy encompasses minimum length, complexity, and age before needing to be changed. Similarly, access controls for restricting access to administrator privileges and files can also be done through Active Directory. This is done through an Access Control List and manages permissions on a user-by-user basis to ensure that they only have access to what they need to in order to do their jobs.

It is important to protect data, especially when it's being transmitted over the internet. To ensure that the data remains confidential, maintains its integrity and non-repudiation, it has to be encrypted using standards such as TLS 1.2. To use this encryption, all that needs to be done is implement it on servers and ensure that the certificate is properly and timely renewed before it expires. It is critically important to keep the private keys confidential and revoke certificate access when it's no longer needed by a client.

Personal Reflection

During the course of completing this assignment, we have learned many aspects of operational technology and the challenges that can arise during efforts to protect it from malicious actors. We had to come up with solutions for these problems that would ensure the systems remained secure while available to use at all times.

These findings apply to the security of real-world OT systems in critical infrastructure due to the fact that the real-world OT system developers and engineers have to consider these cybersecurity challenges as well during the development process of the technology, leading to design considerations and making sure security is in lockstep with the design process.

We could further improve the security of our system and expand its capabilities by considering cybersecurity standards and principles at every step of the development process of our operational technology and use it to build a safer piece of technology.

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